

What is Machine Learning?

Machine Learning (ML) is a part of Artificial Intelligence where computers learn from data instead of being explicitly programmed.

Simple definition:

“Machine learning is teaching computers to learn patterns from data and make decisions.”

Real-life examples:

- YouTube recommending videos
- Netflix suggesting movies
- Google Maps predicting traffic
- Spam email detection

Example:

Instead of writing rules like:

```
if email contains "win money" → spam
```

Machine Learning learns automatically from thousands of emails.

Why Should You Learn Machine Learning?

Since you want to build a career in the coding field, ML is a very powerful skill.

1. High Demand Career

- Data Scientist
- ML Engineer
- AI Engineer

These roles are highly paid.

2. Real-World Impact

ML is used in:

- Healthcare (disease prediction)
- Finance (fraud detection)
- E-commerce (recommendations)

3. Future Technology

ML is the backbone of:

- Chatbots (like ChatGPT)
- Self-driving cars
- Face recognition

4. Smart Problem Solving

Instead of writing complex logic, ML:
Learns patterns automatically

What is Scikit-learn?

Scikit-learn is a Python library used to build machine learning models easily.

It is one of the most popular ML libraries for beginners.

What can you do with Scikit-learn?

- Classification (Spam detection)
- Regression (Price prediction)
- Clustering (Customer grouping)
- Data preprocessing

Example (Very Simple)

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

X = [[1], [2], [3]]
y = [2, 4, 6]

model.fit(X, y)

print(model.predict([[4]])) # Output: 8
```

Here the model learned:

$$y = 2x$$

How Much Maths is Required for Machine Learning?

Beginner level (enough to start ML):

- Basic Algebra
- Basic Statistics
- Basic Probability
- Very basic Linear Algebra

Good news:

You do NOT need advanced mathematics (like heavy calculus proofs) in the beginning.

Maths Concepts You Should Learn (Step-by-Step)

1. Algebra (Most Important)

- Variables (x, y)
- Equations
- Linear equations
- Graph understanding

ML models learn relationships like these.

2. Basic Statistics

- Mean (average)
- Median
- Mode
- Variance
- Standard Deviation

Used to understand data in ML.

3. Probability (Important for ML logic)

- Probability basics
- Conditional probability
- Independent events

Used in predictions and spam detection.

4. Basic Linear Algebra

- Vectors (list of numbers)
- Matrices (table of numbers)

Data in ML is often represented in matrix form.

Practice Questions (with Answers)

If you can solve these → You are ready to start ML 🚀

Algebra Questions

Q1:

If $y = 2x + 3$, find y when $x = 5$

Answer:

$$y = 2(5) + 3 = 13$$

Q2:

Solve: $2x = 10$

Answer:

$$x = 5$$

Statistics Questions

Q3:

Find mean of: 2, 4, 6, 8

Answer:

$$\text{Mean} = (2 + 4 + 6 + 8) / 4 = 5$$

Q4:

Find median of: 1, 3, 5, 7, 9

Answer:

$$\text{Median} = 5$$

Probability Questions

Q5:

What is the probability of getting heads in a coin toss?

Answer:

$$1/2 = 0.5$$

Q6:

A bag has 3 red and 2 blue balls.

What is the probability of picking a red ball?

Answer:

$$3 / 5 = 0.6$$

Linear Algebra Questions

Q7:

Vector addition:

$$[1, 2] + [3, 4]$$

Answer:

$$[4, 6]$$

Q8:

Find the element in the 1st row and 2nd column of the matrix:

$$\begin{bmatrix} 1 & 5 \\ 2 & 8 \end{bmatrix}$$

Answer:

5

Functions (Very Important for ML)

A **function** is a relationship between input and output.

Example:

If:

- $f(2) = 4$
- $f(3) = 9$

ML models = functions that map input $\rightarrow$ output

Questions

Q1:

If $f(x) = x^2$, find $f(4)$

Answer:

$$f(4) = 16$$

Q2:

If $f(x) = 3x + 2$, find $f(5)$

Answer:

$$= 15 + 2 = 17$$

Graph Understanding (Visualization)

Graphs show how input (x) affects output (y)

Example:

Shape: U-shaped curve

ML uses graphs to:

- understand trends
- visualize predictions

Questions

Q3:

For $y = x^2$, what is y when $x = -2$?

Answer:

$$(-2)^2 = 4$$

Q4:

Is $y = x^2$ linear or non-linear?

Answer:

Non-linear

Loss Function (Core ML Concept)

Concept:

Loss function tells:

"How wrong your model is"

Example (Mean Squared Error):

y = actual value

$\hat{y}$ = predicted value

Questions

Q5:

Actual = 5, Predicted = 3

Find error = $(y - \hat{y})$

Answer:

$$= 5 - 3 = 2$$

Q6:

Square error?

Answer:

$$2^2 = 4$$

Gradient (Basic Idea Only)

Gradient = **slope of a function**

Example:

$$\text{Slope} = 2$$

ML uses gradient to:

- reduce error
- improve model

Questions

Q7:

What is slope of $y = 5x$?

Answer:

5

Q8:

If slope is positive, graph goes?

Answer:

Upward

Distance (Used in KNN Algorithm)

Concept:

Distance helps measure similarity

Formula:

Questions

Q9:

Find distance between (1,2) and (4,6)

Answer:

$$= \sqrt{[(4-1)^2 + (6-2)^2]}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} = 5$$

10. Feature & Target (ML Terminology)

- **Feature (X)** → input
- **Target (y)** → output

Example:

Hours Study	Marks
2	30

Feature = Hours

Target = Marks

Questions

Q10:

In house price prediction:

- Size of house = ?

Answer:

Feature

Q11:

Price of house = ?

Answer:

Target